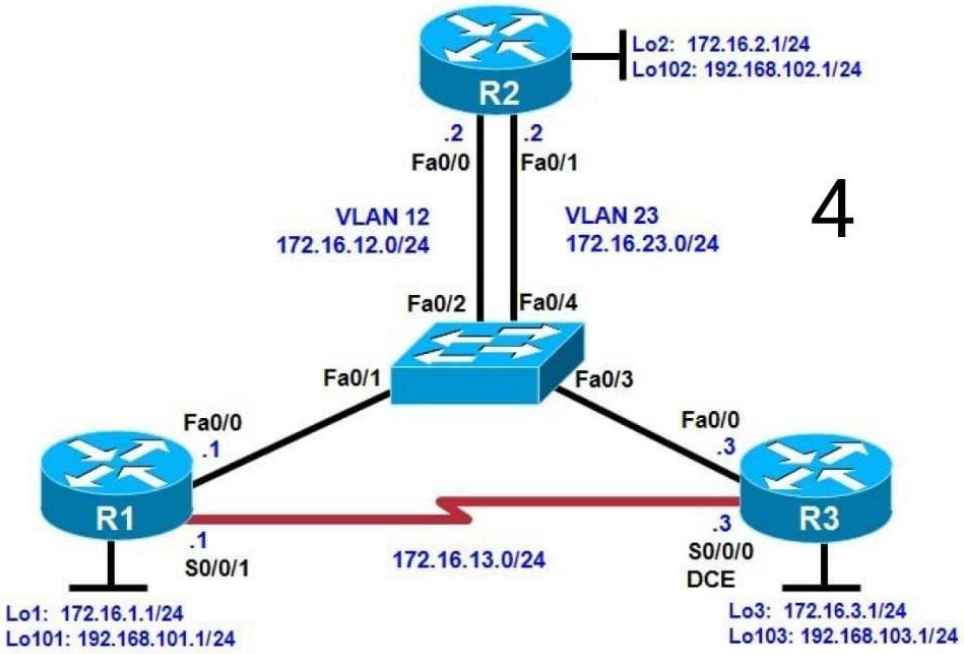
4prac: Manipulating Administrative Distances



● R1 Configuration

```
enable
configure terminal

interface loopback 1
 ip address 172.16.1.1 255.255.255.0
 exit

interface loopback 101
 ip address 192.168.101.1 255.255.255.0
 exit

interface fastethernet 0/0
 ip address 172.16.12.1 255.255.255.0
 no shutdown
 exit

interface serial 0/0/1
 bandwidth 64
 ip address 172.16.13.1 255.255.255.0
 no shutdown
 exit
```

● R2 Configuration

```
enable
configure terminal

interface loopback 2
 ip address 172.16.2.1 255.255.255.0
 exit

interface loopback 102
 ip address 192.168.102.1 255.255.255.0
 exit

interface fastethernet 0/0
 ip address 172.16.12.2 255.255.255.0
 no shutdown
 exit

interface fastethernet 0/1
 ip address 172.16.23.2 255.255.255.0
 no shutdown
 exit
```

● R3 Configuration

```
enable
configure terminal

interface loopback 3
 ip address 172.16.3.1 255.255.255.0
 exit

interface loopback 103
 ip address 192.168.103.1 255.255.255.0
 exit

interface fastethernet 0/0
 ip address 172.16.23.3 255.255.255.0
 no shutdown
 exit

interface serial 0/0/0
 bandwidth 64
 ip address 172.16.13.3 255.255.255.0
 clock rate 64000
 no shutdown
 exit
```

Step 2: Configure Switch VLANs

```
Switch(config)# vlan 12
Switch(config-vlan)# name R1-R2
```

```
Switch(config)# vlan 23
Switch(config-vlan)# name R2-R3
exit
```

Configure Switch Interfaces

```
interface fastEthernet 0/1
 description To R1 Fa0/0
 switchport mode access
 switchport access vlan 12
 exit

interface fastEthernet 0/2
 description To R2 Fa0/0
 switchport mode access
 switchport access vlan 12
 exit

interface fastEthernet 0/3
 description To R3 Fa0/0
 switchport mode access
 switchport access vlan 23
 exit

interface fastEthernet 0/4
 description To R2 Fa0/1
 switchport mode access
 switchport access vlan 23
 exit
```

Step 3: Configure RIP

● R1

```
router rip
 version 2
 no auto-summary
 network 172.16.0.0
 network 192.168.101.0
```

● R2

```
router rip
 version 2
 no auto-summary
 network 172.16.0.0
 network 192.168.102.0
```

● R3

```
router rip
 version 2
 no auto-summary
```

```
 network 172.16.0.0
 network 192.168.103.0
```

Verification

```
show ip route rip
```

Step 4: Configure OSPF

● R1

```
interface loopback 1
 ip ospf network point-to-point
 exit

interface loopback 101
 ip ospf network point-to-point
 exit

router ospf 1
 network 172.16.0.0 0.0.255.255 area 0
 network 192.168.101.0 0.0.0.255 area 0
```

● R2

```
interface loopback 2
 ip ospf network point-to-point
 exit

interface loopback 102
 ip ospf network point-to-point
 exit

router ospf 1
 network 172.16.0.0 0.0.255.255 area 0
 network 192.168.102.0 0.0.0.255 area 0
```

● R3

```
interface loopback 3
 ip ospf network point-to-point
 exit

interface loopback 103
 ip ospf network point-to-point
 exit

router ospf 1
 network 172.16.0.0 0.0.255.255 area 0
 network 192.168.103.0 0.0.0.255 area 0
```

Verification

```
show ip ospf neighbor
show ip route
```

Step 5: Modify Routing Protocol Distance

● R1

```
router rip
 distance 100
```

● R2

```
router rip
 distance 100
```

● R3

```
router rip
 distance 100
```

Verification

```
show ip route
show ip protocols
```

Step 6: Modify Distance Based on Route Source

● R1

```
router ospf 1
 distance 85 192.168.100.0 0.0.3.255
```

● R2

```
router ospf 1
 distance 85 192.168.100.0 0.0.3.255
```

● R3

```
router ospf 1
 distance 85 192.168.100.0 0.0.3.255
```

Verification

```
show ip route
show ip protocols
```

Step 7: Modify Distance Based on an Access List

● R1

```
access-list 1 permit 172.16.0.0 0.0.255.255

router rip
 distance 65 0.0.0.0 255.255.255.255 1
```

● R2

```
access-list 1 permit 172.16.0.0 0.0.255.255

router rip
 distance 65 0.0.0.0 255.255.255.255 1
```

● R3

```
access-list 1 permit 172.16.0.0 0.0.255.255

router rip
 distance 65 0.0.0.0 255.255.255.255 1
```

● Final Verification Commands

Run on any router:

```
show ip protocols
show ip route
show ip route rip
show ip ospf neighbor
```

chatgpt format

R1 Router

■ Copy-paste this in R1

```
enable
configure terminal

interface loopback 1
ip address 172.16.1.1 255.255.255.0
exit

interface loopback 101
ip address 192.168.101.1 255.255.255.0
exit

interface fastethernet 0/0
ip address 172.16.12.1 255.255.255.0
no shutdown
exit

interface serial 0/0/1
bandwidth 64
ip address 172.16.13.1 255.255.255.0
no shutdown
exit

router rip
version 2
no auto-summary
network 172.16.0.0
network 192.168.101.0
distance 100
exit

interface loopback 1
ip ospf network point-to-point
exit

interface loopback 101
ip ospf network point-to-point
exit

router ospf 1
network 172.16.0.0 0.0.255.255 area 0
network 192.168.101.0 0.0.0.255 area 0
distance 85 192.168.100.0 0.0.3.255
exit
```

```
access-list 1 permit 172.16.0.0 0.0.255.255

router rip
distance 65 0.0.0.0 255.255.255.255 1
end
```

R2 Router

■ Copy-paste this in R2

```
enable
configure terminal

interface loopback 2
ip address 172.16.2.1 255.255.255.0
exit

interface loopback 102
ip address 192.168.102.1 255.255.255.0
exit

interface fastethernet 0/0
ip address 172.16.12.2 255.255.255.0
no shutdown
exit

interface fastethernet 0/1
ip address 172.16.23.2 255.255.255.0
no shutdown
exit

router rip
version 2
no auto-summary
network 172.16.0.0
network 192.168.102.0
distance 100
exit

interface loopback 2
ip ospf network point-to-point
exit

interface loopback 102
ip ospf network point-to-point
exit

router ospf 1
network 172.16.0.0 0.0.255.255 area 0
network 192.168.102.0 0.0.0.255 area 0
distance 85 192.168.100.0 0.0.3.255
exit

access-list 1 permit 172.16.0.0 0.0.255.255

router rip
```

```
distance 65 0.0.0.0 255.255.255.255 1
end
```

R3 Router

■ Copy-paste this in R3

```
enable
configure terminal

interface loopback 3
ip address 172.16.3.1 255.255.255.0
exit

interface loopback 103
ip address 192.168.103.1 255.255.255.0
exit

interface fastethernet 0/0
ip address 172.16.23.3 255.255.255.0
no shutdown
exit

interface serial 0/0/0
bandwidth 64
ip address 172.16.13.3 255.255.255.0
clock rate 64000
no shutdown
exit

router rip
version 2
no auto-summary
network 172.16.0.0
network 192.168.103.0
distance 100
exit

interface loopback 3
ip ospf network point-to-point
exit

interface loopback 103
ip ospf network point-to-point
exit

router ospf 1
network 172.16.0.0 0.0.255.255 area 0
network 192.168.103.0 0.0.0.255 area 0
distance 85 192.168.100.0 0.0.3.255
exit

access-list 1 permit 172.16.0.0 0.0.255.255

router rip
distance 65 0.0.0.0 255.255.255.255 1
end
```

Switch Configuration

■ Copy-paste in Switch

```
enable
configure terminal

vlan 12
name R1-R2
exit

vlan 23
name R2-R3
exit

interface fastethernet 0/1
description To R1 Fa0/0
switchport mode access
switchport access vlan 12
exit

interface fastethernet 0/2
description To R2 Fa0/0
switchport mode access
switchport access vlan 12
exit

interface fastethernet 0/3
description To R3 Fa0/0
switchport mode access
switchport access vlan 23
exit

interface fastethernet 0/4
description To R2 Fa0/1
switchport mode access
switchport access vlan 23
exit

end
```

Verification Commands (Run on any router)

■ Copy if needed

```
show ip protocols
show ip route
show ip route rip
show ip ospf neighbor
```